

[Paper 943, On Reciprocants and Differential Invariants, continued from p. 399.]

Going a step further, we form in this manner the expression

$$E = a^{m-1}c + (m-1)a^{m-2}(bd + \frac{1}{2}c^2) + \frac{1}{2}(m-1)(m-2)[a^{m-3}.3b^2c + (m-3)a^{m-4}b.b^3.\frac{1}{6}],$$

that is,

$$\begin{aligned} E &= a^{m-1}c \\ &+ a^{m-3}[(m-1)bd + \frac{1}{2}(m-1)c^2] \\ &+ a^{m-3}.\frac{1}{2}(m-1)(m-2)b^2c \\ &+ a^{m-4}.\frac{1}{24}(m-1)(m-2)(m-3)b^4, \end{aligned}$$

which is sufficient to explain the rule.

It may be noticed that, the operator being linear in μ, ν , we have

$$(\mu, \nu; m, n) = \mu(1, 0; m, n) + \nu(0, 1; m, n).$$

The Alternant of two multilinear operators.

If P, Q are the two operators, we have as usual

$$P.Q = PQ + P*Q,$$

where $P.Q$ denotes the successive operation first with Q and then with P upon any operand, PQ is the mere algebraical product of the operators, and $P*Q$ is the operator obtained by the operation of P upon Q .

Similarly,

$$Q.P = QP + Q*P,$$

and since QP is the same thing as PQ , we obtain

$$P.Q - Q.P = (P*Q) - (Q*P),$$

either of which equal expressions, or say rather the second of them, is called the alternant of P, Q and is written $[P.Q]$: viz. as the definition of the alternant, we have

$$[P.Q] = (P*Q) - (Q*P).$$

We have the remarkable theorem, that the alternant of any two operators $(\mu', \nu'; m', n')$, $(\mu, \nu; m, n)$ is an operator $(\mu_1, \nu_1; m_1, n_1)$, where

$$\begin{aligned} \mu_1 &= (m + m' - 1) \left\{ \frac{\mu'}{m'}(\mu + n'\nu) - \frac{\mu}{m}(\mu' + n\nu') \right\}, \\ \nu_1 &= (n' - n)\nu\nu' + \frac{m'-1}{m'}\mu'\nu - \frac{m-1}{m}\mu\nu', \\ m_1 &= m' + m - 1, \\ n_1 &= n' + n, \end{aligned}$$

or since m_1, n_1 have such simple expressions, say it is an operator $(\mu_1, \nu_1; m' + m - 1, n' + n)$, where μ_1, ν_1 have the values just written down.

We see at once how these values $m_1 = m' + m - 1$, $n_1 = n' + n$, arise: $Q = (\mu, \nu; m, n)$ contains the letters $(a, b, c, d, \dots)$ in the degree m , and it contains differential symbols ∂ which, operating on any function of these letters, diminish the degree by 1; similarly $P = (\mu', \nu'; m', n')$ contains the letters $(a, b, c, d, \dots)$ in the degree m' , and it contains the differential symbols ∂ which, operating on any function of the letters, diminish the degree by 1. Hence $P*Q$ contains the letters in the degree $(m' - 1) + m$, and similarly $Q*P$ contains them in the same degree $(m - 1) + m'$: thus in the alternant the degree is $m' + m - 1$. Again, in Q the weights of the successive functions $A, B, C, \dots$ are $0, 1, 2, \dots$ and these are combined with differential symbols $\partial_{a_n}, \partial_{a_{n+1}}, \dots$ which operating on any function of the letters diminish the weight by $n, n + 1, \dots$

respectively: that is, the terms $A\partial_{a_n}, B\partial_{a_{n+1}}, \dots$ each diminish the weight by n . So in $P = (\mu', \nu'; m', n')$ the weights of the successive terms $A', B', C', \dots$ are $0, 1, 2, \dots$ and these are combined with differential symbols $\partial_{a_{n'}}, \partial_{a_{n'+1}}, \dots$ which operating on any function of the letters diminish the weights by $n', n'+1, \dots$ respectively. We may say that Q is a sum of terms such as $\Theta_k \partial_{a_{n+k}}$, where the subscript k of Θ denotes the weight of Θ ; operating hereon with P , the corresponding term of $P * Q$ is of the form $\Theta_{k-r} \partial_{a_{n+k}}$, or (what is the same thing) the general form of the terms of $P * Q$ is $\Theta_k \partial_{a_{n+n'+k}}$; and in like manner this is the general form of the terms of $Q * P$; and it is thus also the general form of the terms of the alternant $[P, Q]$. It thus appears that, admitting the alternant to be an operator $(\mu_1, \nu_1; m_1, n_1)$, the value of n_1 is $n' + n$.

As an instance of the theorem, we may take

$$[(1, 0; 1, 1), (1, 0; 2, 1)] = (1, 0; 2, 2), \quad b\partial_c$$

we have here

$$\begin{aligned} (1, 0; 1, 1) * (1, 0; 2, 1) - (1, 0; 2, 1) * (1, 0; 1, 1), \\ = a\partial_b * a\partial_c - a\partial_c * a\partial_b \end{aligned}$$

[Detailed bracket computation; the operator $a\partial_b$ acting upon $a\partial_c + b\partial_d + (ac + \frac{1}{2}b^2)\partial_e + (ad + bc)\partial_f + \dots$ yields by Leibniz, term-by-term differentiation, and gathering of like terms, the operator $(1, 0; 2, 2)$. The verification proceeds as detailed in the printed page.]

($= 1, 0; 2, 2$), as it should be.

For the general proof, writing for a moment

$$C_0 = \mu A_0, \quad C_1 = (\mu + \nu) A_1, \quad C_2 = (\mu + 2\nu) A_2, \quad \&c.,$$

and similarly

$$C'_0 = \mu' A'_0, \quad C'_1 = (\mu' + \nu') A'_1, \quad C'_2 = (\mu' + 2\nu') A'_2, \quad \&c.,$$

where the accented symbols refer to the values $(\mu', \nu'; m', n')$, we have

$$\begin{aligned} [(\mu', \nu'; m', n') (\mu, \nu; m, n)] &= C'_0 \partial_{a_{n'}} \bullet C_0 \partial_{a_n} - C_0 \partial_{a_n} \bullet C'_0 \partial_{a_{n'}} \\ &\quad + C'_1 \partial_{a_{n'+1}} + C_1 \partial_{a_{n+1}} + C_2 \partial_{a_{n+2}} \\ &\quad + C'_2 \partial_{a_{n'+2}} + C'_2 \partial_{a_{n'+2}} + C_2 \partial_{a_{n+2}} \\ &\quad \vdots \\ &\quad + C'_n \partial_{a_{n'+n}} + C_{n'} \partial_{a_{n+n'}} \\ &\quad + C'_{n+1} \partial_{a_{n'+n+1}} + C_{n'+1} \partial_{a_{n+n'+1}} \end{aligned}$$

or observing that in the series $C_0, C_1, C_2, \dots$, the first term that contains $a_{n'}$ is $C_{n'}$, and the like as regards the series $C'_0, C'_1, C'_2, \dots$, this is

$$\begin{aligned} &= [(C'_0 C_{n'}) C'_0 - (C_0 C_n) C'_0] \partial_{a_{n+n'}} \\ &\quad + [(C'_0 \partial_{a_{n'}} + C'_1 \partial_{a_{n'+1}}) C_{n'+1} - (C_0 \partial_{a_n} + C_1 \partial_{a_{n+1}}) C'_{n'+1}] \partial_{a_{n+n'+1}} \\ &\quad + [(C'_0 \partial_{a_{n'}} + C'_1 \partial_{a_{n'+1}} + C'_2 \partial_{a_{n'+2}}) C_{n'+2} - \dots] \partial_{a_{n+n'+2}} \\ &\quad \vdots \end{aligned}$$

and it is thus of the form

$$C''_0 \partial_{a_{n+n'}} + C''_1 \partial_{a_{n+n'+1}} + C''_2 \partial_{a_{n+n'+2}} + \&c.,$$

i.e. the value of n_1 is $n + n'$.

I confine myself to the comparison of the first and second coefficients: substituting for the C 's their expressions in terms of the A 's, we ought to have

$$\begin{aligned} \mu' A'_0 \partial_{a_{n'}} (\mu + n' \nu) A_{n'} - \mu A_0 \partial_{a_n} (\mu' + n \nu') A'_{n'} &= \mu_1 A''_0, \\ (\mu' A'_0 \partial_{a_{n'}} + (\mu' + \nu') A'_1 \partial_{a_{n'+1}}) [\mu + (n' + 1) \nu] A_{n'+1} &= (\mu_1 + \nu_1) A''_1. \end{aligned}$$

Now assuming $m_1 = m' + m - 1$, and attending to the values

$$\begin{aligned} A_0 &= \frac{1}{m} a_0^m, & A'_0 &= \frac{1}{m'} a_0^{m'}, & A''_0 &= \frac{1}{m'+m-1} a_0^{m'+m-1}, \\ A_1 &= a_0^{m-1} a_1, & A'_1 &= a_0^{m'-1} a_1, & A''_1 &= a_0^{m'+m-2} a_1, \end{aligned}$$

we see at once that in the first equation the terms contain each of them the factor $a_0^{m'+m-1}$, and in the second equation they contain each of them the factor $a_0^{m'+m-2} a_1$; omitting these factors, we find

$$\begin{aligned} \frac{\mu'}{m'}(\mu + n'\nu) - \frac{\mu}{m}(\mu' + n\nu') &= \frac{\mu_1}{m' + m - 1}, \\ \frac{m' - 1}{m'}(\mu + n'\nu + \nu)(m - 1) + (\mu' + \nu')(\mu + n'\nu + \nu) - (\mu + \nu)(\mu' + n\nu' + \nu') \\ - (m' + m - 1) \frac{\mu_1}{m' + m - 1}(\mu + n'\nu) - (m' + m - 1) \frac{\nu_1}{m' + m - 1} &= \mu_1 + \nu_1. \end{aligned}$$

The first of these gives

$$\mu_1 = (m' + m - 1) \left\{ \frac{\mu'}{m'}(\mu + n'\nu) - \frac{\mu}{m}(\mu' + n\nu') \right\},$$

which is the value of μ_1 : substituting this in the second equation, we have

$$\begin{aligned} \nu_1 &= \frac{\mu'}{m'}(\mu + n'\nu + \nu)(m - 1) - \frac{\mu}{m}(m' - 1)(\mu' + n\nu' + \nu') \\ &\quad + (\mu' + \nu')(\mu + n'\nu + \nu) - (\mu + \nu)(\mu' + n\nu' + \nu') \\ &\quad - (m' + m - 1) \frac{\mu'}{m'}(\mu + n'\nu) - (m' + m - 1) \frac{\mu}{m}(\mu' + n\nu'). \end{aligned}$$

In the left-hand column, the terms containing ν , $(\mu + n'\nu)$ are

$$\left\{ \frac{\mu'}{m'}(m - 1) + \mu' + \nu' - \frac{\mu'}{m'}(m' + m - 1) \right\},$$

and there are besides the terms $\frac{\mu'}{m'}\nu(m - 1) + \nu(\mu' + \nu')$, hence the left-hand column is

$$= \nu\nu'(n' + 1) + \mu\nu' + \mu'\nu \left(\frac{m-1}{m'} + 1 \right).$$

We have therefore

$$\nu_1 = \nu\nu'(n' + 1) + \mu\nu' + \mu'\nu \left(\frac{m-1}{m'} + 1 \right) - \nu\nu'(n + 1) - \mu\nu' \left(\frac{m-1}{m} + 1 \right) - \mu'\nu,$$

that is,

$$\nu_1 = \nu\nu'(n' - n) + \frac{m'-1}{m'} \mu'\nu - \frac{m-1}{m} \mu\nu'.$$

To complete the proof, it would be of course necessary to compare the remaining terms on the two sides respectively: but in what precedes it is shown that, the form being assumed, the expression for the alternant must of necessity be $(\mu_1, \nu_1; m' + m - 1, n' + n)$, where μ_1, ν_1 have their foregoing values.

Resuming the equation

$$[(\mu', \nu'; m', n')(\mu, \nu; m, n)] = (\mu_1, \nu_1; m_1, n_1),$$

where μ_1, ν_1, m_1, n_1 have the before-mentioned values, then if we herein consider $\mu, \nu, m, n; \mu_1, \nu_1, m_1, n_1$ as given, we can find μ', ν', m', n' .

We have $m_1 = m' + m - 1, n_1 = n' + n$, that is, $m' = m_1 - m + 1, n' = n_1 - n$, and substituting these values in the expressions of μ_1 and ν_1 , we find

$$\begin{aligned} \frac{\mu_1}{m_1} &= \frac{\mu'}{m_1 - m + 1} [\mu + (n_1 - n)\nu] - \frac{\mu}{m}(\mu' + n\nu'), \\ \nu_1 &= (n_1 - 2n)\nu\nu' + \frac{m-1}{m_1 - m + 1} \mu'\nu - \frac{m_1 - m}{m} \mu\nu', \end{aligned}$$

that is,

$$\frac{\mu_1}{m_1} = \left\{ \frac{\mu}{(m_1 - m + 1)} + \frac{\nu(n_1 - n)}{m_1 - m + 1} - \frac{\mu}{m} \right\} \mu' - \frac{\mu n}{m} \nu',$$

$$\nu_1 = \left\{ \frac{(m-1)\nu}{(m_1 - m + 1)} \right\} \mu' + \left\{ (n_1 - 2n)\nu - \frac{m_1 - m}{m} \mu \right\} \nu',$$

which are linear equations for the determination of μ' and ν' .

944.

ON PFAFF-INVARIANTS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XXVI. (1893), pp. 195–205.]

1. The functions which I propose to call Pfaff-invariants present themselves and play a leading part in the memoir, Clebsch, “Ueber das *Pfaffsche Problem*” (Zweite Abhandlung), *Crelle*, t. LXI. (1863), pp. 146–179: but it is interesting to consider them for their own sake as invariants, and in the notation which I have elsewhere used for the functions called Pfaffians. The great simplification effected by this notation is, I think, at once shown by the remark that Clebsch’s expression R , which he defines by the periphrasis “Sei ferner R der *rationale* Ausdruck dessen Quadrat der Determinant der a_{ik} gleich ist” (l.c. p. 149), is nothing else than the Pfaffian $1234 \dots 2n - 1. 2n$, and that its differential coefficients $R_{ik} = \frac{\partial R}{\partial a_{ik}}$ are the Pfaffians obtained from the foregoing by the mere omission of any two symbolic numbers i, k .

2. I call to mind that the symbols 12, 13, &c., made use of are throughout such that $12 = -21$, &c.; and that the definition of the successive Pfaffians 12, 1234, &c., is as follows:

$$12 = 12,$$

$$1234 = 12.34 + 13.42 + 14.23,$$

$$123456 = 12.3456 + 13.4562 + 14.5623 + 15.6234 + 16.2345,$$

in which last expression 3456 denotes the Pfaffian $34.56 + 35.64 + 36.45$, and similarly 4562 &c.; and so on for any even number of symbols. Of course, instead of the symbolic numbers 1, 2, 3, &c., we may have any other numbers (0 is frequently used in the sequel as a symbolic number), or we may have letters or other symbols.

3. I use also a function very analogous to a Pfaffian, which is expressed in the same notation, viz. this is

$$\phi\psi 12 = \phi\psi 12,$$

$$\phi\psi 1234 = \phi\psi 12.34 + \phi\psi 13.42 + \phi\psi 14.23,$$

$$+ \phi\psi 34.12 + \phi\psi 42.13 + \phi\psi 23.14,$$

$$\phi\psi 123456 = \phi\psi 12.34.56 + \phi\psi 34.12.56 + \phi\psi 56.12.34 + \&c.,$$

viz. taking any term 12.34.56 of the Pfaffian 123456, $\phi\psi$ is connected successively with each of the binary symbols 12, 34, 56 of the term, so as to give rise to terms containing the quaternary symbols $\phi\psi 12$, &c. Such function may be called a co-Pfaffian.

4. To avoid suffixes I use different letters (x, y) , (x, y, z) , &c., as the case may be, associating these with the numbers (1, 2), (1, 2, 3), &c. In the case of a differential of an even number $2n$ of terms, for instance $Xdx + Ydy + Zdz + Wdw$, I consider the functions 1234, $\phi 1234$, and $\phi\psi 1234$, the first and second of which are Pfaffians, the last a co-Pfaffian, as explained above. To fix in connexion with the differential $Xdx + Ydy + Zdz + Wdw$ the meanings of these expressions, I assume

$$12 = \frac{dX}{dy} - \frac{dY}{dx}, \quad 13 = \frac{dX}{dz} - \frac{dZ}{dx}, \quad \&c.,$$

(of course these imply $12 = -21$, &c.),

$$01 = -10 = X, \quad 02 = -20 = Y, \quad \&c.;$$

ϕ is an arbitrary function of x, y, z, w , and I write

$$\phi 0 = -0\phi = 0, \quad \phi 1 = -1\phi = \frac{d\phi}{dx}, \quad \phi 2 = -2\phi = \frac{d\phi}{dy}, \quad \&c.;$$

ψ is also an arbitrary function of x, y, z, w , and I write

$$\phi\psi 12 = \frac{d\phi}{dx} \frac{d\psi}{dy} - \frac{d\phi}{dy} \frac{d\psi}{dx} = \frac{\partial(\phi, \psi)}{\partial(x, y)}, \quad \&c.$$

(this implies $\phi\psi 21 = -\phi\psi 12$, &c.).

5. Thus, at full length, the functions are

$$\begin{aligned} 1234 &= 12.34 + 13.42 + 14.23 \\ &= \left(\frac{dX}{dy} - \frac{dY}{dx}\right)\left(\frac{dZ}{dw} - \frac{dW}{dz}\right) + \left(\frac{dX}{dz} - \frac{dZ}{dx}\right)\left(\frac{dW}{dy} - \frac{dY}{dw}\right) + \left(\frac{dX}{dw} - \frac{dW}{dx}\right)\left(\frac{dY}{dz} - \frac{dZ}{dy}\right), \end{aligned}$$

$$\phi 1234 = \phi 0.1234 + \phi 1.2340 + \phi 2.3401 + \phi 3.4012 + \phi 4.0123$$

$$\begin{aligned} &= \frac{d\phi}{dx}(23.40 + 24.03 + 20.34) \\ &\quad + \frac{d\phi}{dy}(34.01 + 30.14 + 31.40) \\ &\quad + \frac{d\phi}{dz}(40.12 + 41.20 + 42.01) \\ &\quad + \frac{d\phi}{dw}(01.23 + 02.31 + 03.12) \end{aligned}$$

$$\begin{aligned} \phi 1234 &= \frac{d\phi}{dx} \left\{ -W \left(\frac{dY}{dz} - \frac{dZ}{dy} \right) + Z \left(\frac{dY}{dw} - \frac{dW}{dy} \right) - Y \left(\frac{dZ}{dw} - \frac{dW}{dz} \right) \right\} + \frac{d\phi}{dy} \left\{ X \left(\frac{dZ}{dw} - \frac{dW}{dz} \right) - Z \left(\frac{dX}{dw} - \frac{dW}{dx} \right) - W \left(\frac{dZ}{dx} - \frac{dX}{dz} \right) \right\} \\ &\quad + \frac{d\phi}{dz} \left\{ -W \left(\frac{dX}{dy} - \frac{dY}{dx} \right) - Y \left(\frac{dW}{dx} - \frac{dX}{dw} \right) + X \left(\frac{dW}{dy} - \frac{dY}{dw} \right) \right\} + \frac{d\phi}{dw} \left\{ X \left(\frac{dY}{dz} - \frac{dZ}{dy} \right) + Y \left(\frac{dZ}{dx} - \frac{dX}{dz} \right) + Z \left(\frac{dX}{dy} - \frac{dY}{dx} \right) \right\}. \end{aligned}$$

$$\begin{aligned} \phi\psi 1234 &= \phi\psi 12.34 + \phi\psi 13.42 + \phi\psi 14.23 + \phi\psi 34.12 + \phi\psi 42.13 + \phi\psi 23.14 \\ &= \frac{\partial(\phi, \psi)}{\partial(x, y)} \left(\frac{dZ}{dw} - \frac{dW}{dz} \right) + \frac{\partial(\phi, \psi)}{\partial(x, z)} \left(\frac{dW}{dy} - \frac{dY}{dw} \right) + \frac{\partial(\phi, \psi)}{\partial(x, w)} \left(\frac{dY}{dz} - \frac{dZ}{dy} \right) + \frac{\partial(\phi, \psi)}{\partial(z, w)} \left(\frac{dX}{dy} - \frac{dY}{dx} \right) + \frac{\partial(\phi, \psi)}{\partial(y, w)} \left(\frac{dZ}{dx} - \frac{dX}{dz} \right) + \frac{\partial(\phi, \psi)}{\partial(y, z)} \left(\frac{dW}{dx} - \frac{dX}{dw} \right). \end{aligned}$$

6. The invariantive property of the functions consists herein, viz. if we have

$$Xdx + Ydy + Zdz + Wdw = Pdp + Qdq + Rdr + Sds,$$

so that p, q, r, s , and thence also P, Q, R, S are functions each of them of x, y, z, w , then we have

$$\begin{aligned} 1234 \frac{\partial(x, y, z, w)}{\partial(p, q, r, s)} &= (1234)' \frac{\partial(p, q, r, s)}{\partial(x, y, z, w)}, \\ \phi 1234 \frac{\partial(x, y, z, w)}{\partial(p, q, r, s)} &= (\phi 1234)' \frac{\partial(p, q, r, s)}{\partial(x, y, z, w)}, \\ \phi\psi 1234 \frac{\partial(x, y, z, w)}{\partial(p, q, r, s)} &= (\phi\psi 1234)' \frac{\partial(p, q, r, s)}{\partial(x, y, z, w)}, \end{aligned}$$

where the accented functions refer to (p, q, r, s, P, Q, R, S) , and where for greater symmetry I have separated the symbolical numerator and denominator $\partial(p, q, r, s)$ and $\partial(x, y, z, w)$; each of these equations really contains

$$\frac{\partial(p, q, r, s)}{\partial(x, y, z, w)},$$

which is the functional determinant $\frac{\partial(p, q, r, s)}{\partial(x, y, z, w)}$ in regard to (x, y, z, w) : or, if we please, it contains the reciprocal hereof

$$\frac{\partial(x, y, z, w)}{\partial(p, q, r, s)},$$

which is the functional determinant $\frac{\partial(x, y, z, w)}{\partial(p, q, r, s)}$ in regard to (p, q, r, s) .

7. The equations give

$$\frac{\phi 1234}{1234} = \frac{(\phi 1234)'}{(1234)'},$$

$$\frac{\phi \psi 1234}{1234} = \frac{(\phi \psi 1234)'}{(1234)'},$$

and then the expressions on the left-hand are absolute invariants in respect to the transformation of

$$Xdx + Ydy + Zdz + Wdw \text{ into } Pdp + Qdq + Rdr + Sds.$$

They are, in fact, (for $2n = 4$) Clebsch's derivatives (ϕ) and (ϕ, ψ) .

8. For the Pfaffian reduction

$$Xdx + Ydy + Zdz + Wdw = Fdf + Gdg,$$

we may write

$$P, Q, R, S = F, G, 0, 0,$$

$$p, q, r, s = f, g, F, G,$$

viz. we take f, g, F, G as the new independent variables; we thus have

$$01' = F, \quad 02' = G, \quad 03' = 0, \quad 04' = 0,$$

$$12' = 0, \quad 13' = 1, \quad 14' = 0, \quad 23' = 0, \quad 24' = 1, \quad 34' = 0,$$

$$(1234)' = 12' \cdot 34' + 13' \cdot 42' + 14' \cdot 23' = -1;$$

and similarly

$$(\phi 1234)' = - \left\{ F \frac{d\phi}{dF} + G \frac{d\phi}{dG} \right\},$$

$$(\phi \psi 1234)' = - \left\{ \frac{\partial(\phi, \psi)}{\partial(f, F)} + \frac{\partial(\phi, \psi)}{\partial(g, G)} \right\},$$

where, in the equations, the $-$ sign presents itself by reason that $2n, = 4$, is the double of an even number, or say that n is even; in the case of $2n$, the double of an odd number, that is, n odd, the sign would have been $+$.

9. We thus have

$$(\phi) = \frac{\phi 1234}{1234} = F \frac{d\phi}{dF} + G \frac{d\phi}{dG},$$

$$(\phi, \psi) = \frac{\phi \psi 1234}{1234} = \frac{\partial(\phi, \psi)}{\partial(f, F)} + \frac{\partial(\phi, \psi)}{\partial(g, G)},$$

and in particular, by giving to ϕ and ψ the values f, g, F, G , we find

$$(f) = 0, \quad (g) = 0, \quad (F) = F, \quad (G) = G,$$

$$(f, g) = 0, \quad (f, F) = -F, \quad (f, G) = 0, \quad (g, F) = 0, \quad (g, G) = -G, \quad (F, G) = 1,$$

which are Clebsch's equations; in the case of $2n$ terms, the number of them is

$$n + n + \frac{1}{2}(n^2 - n) + \frac{1}{2}(n^2 - n) + n^2 = 2n + n^2 - n + n^2,$$

$$= n(2n + 1), \text{ or } \frac{1}{2} \cdot 2n(2n + 1), \text{ as it should be.}$$

10. It may be remarked that we have

$$Fdf + Gdg = Fd\left(f + \frac{G}{F}g\right) - Fgd\frac{G}{F},$$

or writing this $= F'df' + G'dg'$, we have

$$F' = F, \quad f' = f + \frac{Gg}{F}, \quad G' = Fg, \quad g' = \frac{G}{F},$$

whence conversely

$$F = F', \quad f = f' - \frac{G'g'}{F'}, \quad G = F'g', \quad g = \frac{G'}{F'}, \quad (Gg = -G'g').$$

The ten equations $(f) = 0$, $(g) = 0$, &c., ought then to lead to the corresponding ten equations $(f') = 0$, $(g') = 0$, &c., and it is easy to verify that they do so; for instance, we have

$$(f') = \left(f + \frac{Gg}{F}\right) = (f) + \frac{1}{F} g (G) + g \left(\frac{G}{F}\right),$$

where

$$g \left(\frac{G}{F}\right) = \frac{g}{F}(G) - \frac{Gg}{F^2}(F), = 0,$$

and thus $(f') = 0$. And again,

$$(f', g') = \left(f + \frac{Gg}{F}, \frac{G}{F}\right) = \left(f, \frac{G}{F}\right) + \left(\frac{Gg}{F}, \frac{G}{F}\right) = \left(f, \frac{G}{F}\right) + \left(\frac{Gg}{F}, \frac{G}{F}\right),$$

where the last term vanishes; the remaining terms are

$$= \frac{1}{F}(f, G) - \frac{G}{F^2}(f, F), \text{ that is, } -\frac{G}{F} + \frac{G}{F^2} F, = 0.$$

There is, of course, the like transformation

$$Fdf + Gdg = Gd\left(g + \frac{F}{G}f\right) - Gfd\frac{F}{G}.$$

11. I have, for better exhibiting the results, taken $2n = 4$, but the most simple case for an even number of terms is $2n = 2$. Here we have $Xdx + Ydy = Pdp + Qdq$, and the functions to be considered are

$$12 = 12$$

$$\phi 12 = \phi 0 \cdot 12 + \phi 1 \cdot 20 + \phi 2 \cdot 01 = -Y \frac{d\phi}{dx} + X \frac{d\phi}{dy},$$

$$\phi\psi 12 = \phi\psi 12.$$

We have here

$$X = P \frac{dp}{dx} + Q \frac{dq}{dx}, \quad Y = P \frac{dp}{dy} + Q \frac{dq}{dy},$$

and the invariantive properties are easily verified.

12. Thus

$$12 = \frac{dX}{dy} - \frac{dY}{dx}, = \left(\frac{dP}{dy} \frac{dp}{dx} + \frac{dP}{dx} \frac{dp}{dy}\right) + \left(\frac{dQ}{dy} \frac{dq}{dx} - \frac{dQ}{dx} \frac{dq}{dy}\right),$$

or, writing herein

$$\frac{dP}{dx} = \frac{dP}{dp} \frac{dp}{dx} + \frac{dP}{dq} \frac{dq}{dx},$$

and the like values for $\frac{dP}{dy}$ and for $\frac{dQ}{dx}$, $\frac{dQ}{dy}$, we have

$$12 = \left(\frac{dP}{dq} - \frac{dQ}{dp}\right) \left(\frac{dp}{dy} \frac{dq}{dx} - \frac{dp}{dx} \frac{dq}{dy}\right) = (12)' \frac{\partial(p,q)}{\partial(x,y)}.$$

Similarly, we find

$$\phi 12 = -Y \frac{d\phi}{dx} + X \frac{d\phi}{dy} = \left(\frac{d\phi}{dp} \frac{dp}{dx} + \frac{d\phi}{dq} \frac{dq}{dx}\right) X + \dots = (\phi 12)' \frac{\partial(p,q)}{\partial(x,y)},$$

and

$$\phi\psi 12 = \frac{d\phi}{dx} \frac{d\psi}{dy} - \frac{d\phi}{dy} \frac{d\psi}{dx} = \left(\frac{d\phi}{dp} \frac{dp}{dq} - \frac{d\phi}{dq} \frac{dp}{dp}\right) \left(\frac{dp}{dx} \frac{dq}{dy} - \frac{dp}{dy} \frac{dq}{dx}\right) = (\phi\psi 12)' \frac{\partial(p,q)}{\partial(x,y)}.$$

We thus have

$$\begin{aligned} 12 \partial(x, y) &= (12)' \partial(p, q), \\ \phi 12 \partial(x, y) &= (\phi 12)' \partial(p, q), \\ \phi\psi 12 \partial(x, y) &= (\phi\psi 12)' \partial(p, q), \end{aligned}$$

or say

$$\frac{\phi_{12}}{12} = \frac{(\phi_{12})'}{(12)'}, \text{ and } \frac{\phi\psi_{12}}{12} = \frac{(\phi\psi_{12})'}{(12)'}$$

The proof is the same in principle for $2n = 4$, or any other even value.

13. The theory is very similar in the case of an odd number $2n + 1$ of terms; thus $2n + 1 = 3$, the forms are

$$0123, \quad \phi 123, \quad \text{and} \quad \phi\psi 0123,$$

the first and second of which are Pfaffians, the third of them co-Pfaffian: the developed expression of this last is

$$\phi\psi 0123 = \phi\psi 01 \cdot 23 + \phi\psi 02 \cdot 31 + \phi\psi 03 \cdot 12,$$

and to fix the meaning hereof we write

$$\phi\psi 01 = 0, \quad \phi\psi 02 = 0, \quad \phi\psi 03 = 0.$$

Hence, the differential expression being $Xdx + Ydy + Zdz$, we have

$$\begin{aligned} 0123 &= 01 \cdot 23 + 02 \cdot 31 + 03 \cdot 12 \\ &= X \left(\frac{dY}{dz} - \frac{dZ}{dy} \right) + Y \left(\frac{dZ}{dx} - \frac{dX}{dz} \right) + Z \left(\frac{dX}{dy} - \frac{dY}{dx} \right), \\ \phi 123 &= \phi 1 \cdot 23 + \phi 2 \cdot 31 + \phi 3 \cdot 12 \\ &= \frac{d\phi}{dx} \left(\frac{dY}{dz} - \frac{dZ}{dy} \right) + \frac{d\phi}{dy} \left(\frac{dZ}{dx} - \frac{dX}{dz} \right) + \frac{d\phi}{dz} \left(\frac{dX}{dy} - \frac{dY}{dx} \right), \\ \phi\psi 0123 &= \phi\psi 23 \cdot 01 + \phi\psi 31 \cdot 02 + \phi\psi 12 \cdot 03 \\ &= X \frac{\partial(\phi, \psi)}{\partial(y, z)} + Y \frac{\partial(\phi, \psi)}{\partial(z, x)} + Z \frac{\partial(\phi, \psi)}{\partial(x, y)}. \end{aligned}$$

14. For the transformation

$$Xdx + Ydy + Zdz = Pdp + Qdq + Rdr,$$

we have

$$\begin{aligned} 0123 \partial(x, y, z) &= (0123)' \partial(p, q, r), \\ \phi 123 \partial(x, y, z) &= (\phi 123)' \partial(p, q, r), \\ \phi\psi 0123 \partial(x, y, z) &= (\phi\psi 0123)' \partial(p, q, r), \end{aligned}$$

and consequently

$$\frac{\phi_{123}}{0123} = \frac{(\phi_{123})'}{(0123)'}, \quad \frac{\phi\psi 0123}{0123} = \frac{(\phi\psi 0123)'}{(0123)'},$$

so that the left-hand functions are absolute invariants.

15. If in particular, $Xdx + Ydy + Zdz = df + Gdg$, then we may write

$$P, Q, R = 1, G, 0,$$

$$p, q, r = f, g, G.$$

Hence

$$01' = 1, \quad 02' = G, \quad 03' = 0; \quad 23' = 1, \quad 31' = 0, \quad 12' = 0,$$

or say

$$(0123)' = 1, \quad (\phi 123)' = \frac{d\phi}{df}, \quad (\phi\psi 0123)' = \frac{\partial(\phi, \psi)}{\partial(g, G)} + G \frac{\partial(\phi, \psi)}{\partial(f, G)},$$

and therefore

$$0123 \partial(x, y, z) = \partial(f, g, G),$$

and we thus have

$$\phi 123 \partial(x, y, z) = \frac{d\phi}{df} \partial(f, g, G),$$

$$\phi\psi 0123 \partial(x, y, z) = \left[\frac{\partial(\phi, \psi)}{\partial(g, G)} + G \frac{\partial(\phi, \psi)}{\partial(f, G)} \right] \partial(f, g, G),$$

and thus

$$(\phi) = \frac{\phi 123}{0123} = \frac{d\phi}{df},$$

$$(\phi, \psi) = \frac{\phi\psi 0123}{0123} = \frac{\partial(\phi, \psi)}{\partial(g, G)} + G \frac{\partial(\phi, \psi)}{\partial(f, G)};$$

viz. we thus have derivatives (ϕ) and (ϕ, ψ) analogous to (but quite different in form from) those of Clebsch in the case of an even number of terms.

In particular, writing $\phi, \psi = f, g, G$, we obtain

$$(f) = 1, \quad (g) = 0, \quad (G) = 0; \quad (f, g) = 0, \quad (f, G) = -G, \quad (g, G) = 1,$$

which are the analogues of Clebsch's formulæ.

16. It is interesting to compare the formulæ for the two cases

$$Xdx + Ydy + Zdz + Wdw = Fdf + Gdg,$$

and

$$Xdx + Ydy + Zdz = df + Gdg.$$

In the former case f and g are symmetrically related to each other, and we may say that ($f = \text{const.}$ and $g = \text{const.}$) is a solution of $Xdx + Ydy + Zdz + Wdw = 0$; we have $(f) = 0$ and $(g) = 0$. In the second case ($f = \text{const.}$ and $g = \text{const.}$) is still a solution of $Xdx + Ydy + Zdz = 0$, but f and g are not symmetrically related to each other, and we have $(f) = 1, (g) = 0$. Moreover, in the first case $(G) = G$, but in the second case $(G) = 0$, an equation of the same form as $(g) = 0$; the reason is that we have

$$Xdx + Ydy + Zdz = df + Gdg, = d(f + Gg) - gdG,$$

so that, besides the solution ($f = \text{const.}$ and $g = \text{const.}$), we have the solution

$$(f + Gg = \text{const. and } G = \text{const.}).$$

17. The remark just made may be further developed: we have

$$Xdx + Ydy + Zdz = df + Gdg, = d(f + Gg) - gdG, = df' + G'dg',$$

suppose; where $f' = f + Gg, G' = -g, g' = G$, and therefore also $f = f' + G'g', G = g', g = -G'$: the equations

$$(f) = 1, \quad (g) = 0, \quad (G) = 0, \quad (f, g) = 0, \quad (f, G) = -G, \quad (g, G) = 1,$$

should lead to

$$(f') = 1, \quad (g') = 0, \quad (G') = 0, \quad (f', g') = 0, \quad (f', G') = -G', \quad (g', G') = 1.$$

There is no difficulty in verifying this; thus the equations $(g) = 0, (G) = 0$, give at once $(g') = 0, (G') = 0$; and then the equation $(f) = 1$ gives $(f' + G'g') = 1$, that is,

$$(f') + g'(G') + G'(g') = 1, \text{ or } (f') = 1.$$

So again $(g, G) = 1$ gives $(g, G') = 1$; and then $(f, g) = 0$ gives $(f' + G'g', G') = 0$, that is,

$$(f', G') + G'(g', G') + g'(G', G') = 0, \text{ or } (f', G') = -G'.$$

And finally, $(f, G) = -G$ gives $(f' + G'g', g') + g' = 0$, that is,

$$(f', g') + G'(g', g') + g'(G', g') + g' = 0, \text{ or } (f', g') = 0.$$

I stop to give the direct verification of the equations $(f) = 1, (g) = 0, (G) = 0$. We have, as above,

$$Xdx + Ydy + Zdz = df + Gdg,$$

that is,

$$X = \frac{df}{dx} + G \frac{dg}{dx}, \quad Y = \frac{df}{dy} + G \frac{dg}{dy}, \quad Z = \frac{df}{dz} + G \frac{dg}{dz},$$

and thence

$$\begin{aligned} 23 &= \frac{dY}{dz} - \frac{dZ}{dy} = \frac{dG}{dz} \frac{dg}{dy} - \frac{dG}{dy} \frac{dg}{dz}, \\ 31 &= \frac{dZ}{dx} - \frac{dX}{dz} = \frac{dG}{dx} \frac{dg}{dz} - \frac{dG}{dz} \frac{dg}{dx}, \\ 12 &= \frac{dX}{dy} - \frac{dY}{dx} = \frac{dG}{dy} \frac{dg}{dx} - \frac{dG}{dx} \frac{dg}{dy}. \end{aligned}$$

Hence, multiplying first by $\frac{df}{dx}$, $\frac{df}{dy}$, $\frac{df}{dz}$, that is,

$$X = \frac{df}{dx} + G \frac{dg}{dx}, \quad Y = \frac{df}{dy} + G \frac{dg}{dy}, \quad Z = \frac{df}{dz} + G \frac{dg}{dz},$$

and adding, we have

$$23 \frac{df}{dx} + 31 \frac{df}{dy} + 12 \frac{df}{dz} = X23 + Y31 + Z12,$$

that is, $\phi 123 = 0123$, or $(f) = 1$.

And then multiplying secondly by $\frac{dg}{dx}$, $\frac{dg}{dy}$, $\frac{dg}{dz}$ and adding, and thirdly by $\frac{dG}{dx}$, $\frac{dG}{dy}$, $\frac{dG}{dz}$ and adding, we obtain

$$23 \frac{dg}{dx} + 31 \frac{dg}{dy} + 12 \frac{dg}{dz} = 0, \text{ that is, } (g) = 0,$$

and

$$23 \frac{dG}{dx} + 31 \frac{dG}{dy} + 12 \frac{dG}{dz} = 0, \text{ that is, } (G) = 0.$$

To exhibit more clearly the formulæ for any odd number of terms, I take $2n + 1 = 5$,

$$Xdx + Ydy + Zdz + Wdw + Tdt = df + Gdg + Hdh.$$

We have here

$$\begin{aligned} (\phi) &= \frac{\phi 12345}{012345} = \frac{d\phi}{df}, \\ (\phi, \psi) &= \frac{\phi\psi 012345}{012345} = \frac{\partial(\phi, \psi)}{\partial(g, G)} + \frac{\partial(\phi, \psi)}{\partial(h, H)} - G \frac{\partial(\phi, \psi)}{\partial(f, G)} - H \frac{\partial(\phi, \psi)}{\partial(f, H)}; \end{aligned}$$

and in particular,

$$\begin{aligned} (f) &= 1; \quad (g) = 0, \quad (h) = 0; \quad (G) = 0, \quad (H) = 0; \\ (f, g) &= 0, \quad (f, h) = 0, \quad (f, G) = -G, \quad (f, H) = -H, \\ (g, h) &= 0, \quad (G, H) = 0, \quad (g, G) = 1, \quad (g, H) = 0, \\ (h, G) &= 0, \quad (h, H) = 1; \end{aligned}$$

in all

$$\begin{aligned} &1 + 2n + 2n + \frac{1}{2}(n^2 - n) + \frac{1}{2}(n^2 - n) + n^2 \\ &= 1 + 4n + n^2 - n + n^2, = 2n^2 + 3n + 1, = \frac{1}{2}(2n + 1)(2n + 2) \end{aligned}$$

equations.

We can, by what precedes, at once express the conditions which must be satisfied in order that a differential expression $X_1dx_1 + X_2dx_2 + \cdots + X_vdx_v$, may be reducible to one of the special forms df , Fdf , $df + F_1df_1$, &c.; viz. if we have

$$\begin{aligned} X_1dx_1 + X_2dx_2 + \cdots + X_vdx_v &= df, \quad \text{then } 12 = 0, \text{ \&c.} \\ &= Fdf, \quad \text{,, } 0123 = 0, \text{ \&c.} \\ &= df + F_1df_1, \quad \text{,, } 1234 = 0, \text{ \&c.} \\ &= Fdf + F_1df_1, \quad \text{,, } 012345 = 0, \text{ \&c.} \\ &\text{\&c.} \end{aligned}$$

where the numbers 12, 1234, 12345, &c., represent any combinations out of the numbers 1, 2, 3, ..., v . Of course, if v is not sufficiently large to furnish such a combination, then there is no condition to be satisfied; thus if

$$X_1dx_1 + X_2dx_2 + X_3dx_3 = df + F_1df_1,$$

there is no condition to be satisfied.

945.

NOTE ON LACUNARY FUNCTIONS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XXVI. (1893), pp. 279–281.]

The present note is founded upon Poincaré's paper "Sur les fonctions à espaces lacunaires," *Amer. Math. Jour.*, t. XIV. (1892), pp. 201–221.

If the complex variable $z = x + iy$ is represented as usual by a point the coordinates of which are (x, y) , and if $U_0, U_1, U_2, \dots$ denote an infinite series of given functions of z , then the equation

$$fz = U_0 + U_1 + U_2 + \dots$$

defines a function of z , but only for those values of z for which the series is convergent, or say for points within a certain region Θ ; and within this region, it defines the successive derived functions $f'z, f''z, f'''z, \dots$

Taking $l, = h + ik$, as an increment of z , we define the function of $z + l$ by the equation

$$f(z + l) = fz + \frac{l}{1} f'z + \frac{l^2}{1.2} f''z + \dots,$$

but only for values of l for which the series is convergent: it may very well be, and it is in general the case, that we thereby extend the definition of fz so as to make it applicable to points within a larger region Θ_1 ; and then considering fz as defined within this larger region Θ_1 , we may pass from it to a still larger region Θ_2 ; and so on indefinitely, or until we cover the whole infinite plane.

For instance, the equation

$$fz = 1 + z + z^2 + z^3 + \dots$$

defines the function $1 \div (1 - z)$ for values of z for which $\text{mod. } z < 1$, that is, for points within the circle $x^2 + y^2 - 1 = 0$; and this being so,

$$f(z + l) = \frac{1}{1-z} + \frac{l}{(1-z)^2} + \frac{l^2}{(1-z)^3} + \dots$$

extends the definition to the larger region for which this is a convergent series: the condition of convergency is

$$\text{mod. } \frac{l}{1-z} < 1, \text{ that is, mod. } \frac{h+ik}{1-x-iy} < 1, \text{ or } h^2 + k^2 < (1-x)^2 + y^2.$$

The condition is that the distance $\sqrt{(h^2 + k^2)}$ must not exceed the distance $\sqrt{\{(1-x)^2 + y^2\}}$ of the point $z = x + iy$, from the point $(x = 1, y = 0)$; the point z is strictly within the circle $x^2 + y^2 = 1$, but taking it on the circumference, the condition is that the point $z + l$ must lie within a circle having its centre on the circle $x^2 + y^2 - 1 = 0$ and passing through the point $(x = 1, y = 0)$. Taking $\cos \theta$ and $\sin \theta$ for the coordinates of the centre, the equation of this circle is

$$(x - \cos \theta)^2 + (y - \sin \theta)^2 = (1 - \cos \theta)^2 + \sin^2 \theta,$$

that is,

$$x^2 + y^2 - 1 - 2 \cos \theta (x - 1) - 2y \sin \theta = 0;$$

and the envelope of these circles is

$$(x^2 + y^2 - 1)^2 - 4(x - 1)^2 - 4y^2 = 0,$$

or, as this may be written,

$$(x^2 + y^2)^2 - 6(x^2 + y^2) + 8x - 3 = 0,$$

or in the form

$$\begin{aligned} (x^2 + y^2 - 3)^2 + 4(2x - 1) &= 0, \\ y^4 + 2y^2(x^2 - 3) + (x - 1)^2(x + 3) &= 0. \end{aligned}$$

The curve is a cuspidal Cartesian. To put this in evidence, observe that the equation may be written

$$-4y^2 + \{(x - 1)^2 + y^2\}\{(x - 1)^2 + y^2 + 4x - 4\} = 0,$$

viz. writing

$$\begin{aligned} A &= x + iy - 1, \\ B &= x - iy - 1, \\ Z &= -1, \end{aligned}$$

then

$$\begin{aligned} A - B &= 2iy, \\ AB &= x^2 - 1 + y^2, \\ A + B &= 2x - 2, \end{aligned}$$

or the equation is

$$Z^2(A - B)^2 + AB\{AB - 2Z(A + B)\} = 0,$$

that is,

$$Z^2A^2 + Z^2B^2 + A^2B^2 - 2Z^2AB - 2ZA^2B - 2ZAB^2 = 0,$$

which is the equation of a bicuspidal quartic curve, having for cusps the vertices of the triangle $A = 0$, $B = 0$, $Z = 0$.

The region within which the function is $= \frac{1}{1-z}$ is thus extended to the area within the Cartesian curve, say this is the region Θ_1 ; starting from this curve instead of the circle (viz. by considering the envelope of the circle having its centre on the curve and passing through the point $x = 1$, $y = 0$), we obtain a second curve, a closed curve, which instead of having a cusp on the axis of x cuts this axis at right angles at a point the distance of which from the origin is greater than 1; and we thus extend the region to the area within this second curve, say this is the region Θ_2 . And proceeding in this way, we ultimately extend the region to the whole of the infinite plane.

But the functions $U_0, U_1, U_2, \dots$ may be such that for every value whatever of l , for which the point $z + l$ is outside the region Θ , the series

$$fz + \frac{l}{1} f'z + \frac{l^2}{1 \cdot 2} f''z + \dots$$

is divergent, and we are in this case unable to define the function fz for points outside the region Θ ; the function then exists only for points inside the region Θ , and for points outside this region it is non-existent; a function such as this, existing only for points within a certain region and not for the whole of the infinite plane, is said to be a lacunary function.

946.

NOTE ON THE THEORY OF ORTHOMORPHOSIS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XXVI. (1893), pp. 282–288.]

The equation of any given curve whatever, $\Omega = 0$, may be expressed in the form

$$\phi(x + iy) + \phi(x - iy) = 0.$$

Let χ be any odd function; then since

$$\phi(x - iy) = -\phi(x + iy),$$

we have

$$\chi\phi(x - iy) = \chi\{-\phi(x + iy)\} = -\chi\phi(x + iy),$$

that is,

$$\chi\phi(x + iy) + \chi\phi(x - iy) = 0.$$

Assuming that Ω is a real function, that is, a function with real coefficients, then also $\phi(x + iy)$ will be a function with real coefficients, or say a real function of $x + iy$; the function χ may be real or imaginary,

but if imaginary, then the i of the coefficients does not change its sign in the passage from $\chi\phi(x + iy)$ to $\chi\phi(x - iy)$.

In proof of the assumed theorem, imagine the equation $\Omega = 0$ expressed as an equation between $x + iy$ and $x - iy$, or, supposing it solved in regard to $x - iy$, take the form of it to be $x - iy = f(x + iy)$: let u_n be a function of n satisfying the equation of differences $u_{n+1} = fu_n$; and let $\phi(x + iy)$ be determined as a function of $x + iy$ by the elimination of n from the equations

$$x + iy = u_n, \quad \phi(x + iy) = \cos n\pi;$$

we thence have

$$x - iy = fu_n, = u_{n+1},$$

and consequently

$$\phi(x - iy) = \cos(n + 1)\pi,$$

that is,

$$\phi(x + iy) + \phi(x - iy) = 0,$$

viz. this equation is a transformation of the equation $\Omega = 0$, and thus it appears that the equation $\Omega = 0$ can always be thrown into the last-mentioned form.

As an example, take the equation $y = ax + b$: which, putting for a moment $\xi = x + iy$, $\eta = x - iy$, is

$$\frac{1}{2i}(\xi - \eta) = \frac{1}{2}a(\xi + \eta) + b,$$

that is,

$$\eta = \frac{i+a}{i-a} \xi + \frac{2b}{i-a};$$

we have therefore

$$u_{n+1} = \frac{i+a}{i-a} u_n + \frac{2b}{i-a},$$

a solution of which is

$$u_n = \left(\frac{i+a}{i-a} \right)^n - \frac{b}{a};$$

putting this = ξ , we have

$$n = \frac{1}{\log \frac{i+a}{i-a}} \log \left(\xi + \frac{b}{a} \right),$$

and thence

$$\phi\xi = \cos \frac{\pi}{\log \frac{i+a}{i-a}} \log \left(\xi + \frac{b}{a} \right),$$

where observe that, writing $a + i = Re^{i\alpha}$ and therefore $a - i = Re^{-i\alpha}$, we have

$$\cos \alpha = \frac{a}{\sqrt{(a^2+1)}}, \quad \sin \alpha = \frac{1}{\sqrt{(a^2+1)}},$$

or say $\cot \alpha = a$, and then

$$\frac{i+a}{i-a} = e^{2\alpha i}, \text{ or } \log \frac{i+a}{i-a} = i(2\alpha + \pi),$$

whence

$$\phi\xi = \cos \frac{\pi}{i(2\alpha+\pi)} \log \left(\xi + \frac{b}{a} \right), = \cos \frac{\pi}{2\alpha+\pi} \log \left(\xi + \frac{b}{a} \right),$$

a real function of ξ .

In verification of the equation $\phi\xi + \phi\eta = 0$, we have

$$\phi\eta = \cos \frac{\pi}{\log \frac{i+a}{i-a}} \log \left(\eta + \frac{b}{a} \right),$$

where

$$\log \left(\eta + \frac{b}{a} \right) = \log \left(\frac{i+a}{i-a} \xi + \frac{2b}{i-a} + \frac{b}{a} \right) = \log \frac{i+a}{i-a} \left(\xi + \frac{b}{a} \right) = \log \frac{i+a}{i-a} + \log \left(\xi + \frac{b}{a} \right),$$

and thence

$$\phi\eta = \cos \frac{\pi}{\log \frac{i+a}{i-a}} \left\{ \log \frac{i+a}{i-a} + \log \left(\xi + \frac{b}{a} \right) \right\}$$

$$= \cos \left\{ \pi + \frac{\pi}{\log \frac{i+a}{i-a}} \log \left(\xi + \frac{b}{a} \right) \right\} = -\cos \frac{\pi}{\log \frac{i+a}{i-a}} \log \left(\xi + \frac{b}{a} \right),$$

that is, $\phi\eta = -\phi\xi$, or $\phi\xi + \phi\eta = 0$, the equation in question.

I remark, in passing, that the same equation $y = ax + b$ might have been put in the form $\phi x + \phi y = 0$, viz. assuming

$$\phi x = \cos \frac{\pi}{\log a} \log \left(x - \frac{b}{1-a} \right),$$

then

$$\begin{aligned} \phi y &= \cos \frac{\pi}{\log a} \log \left(ax + b - \frac{b}{1-a} \right) = \cos \frac{\pi}{\log a} \log a \left(x - \frac{b}{1-a} \right) \\ &= \cos \frac{\pi}{\log a} \left\{ \log a + \log \left(x - \frac{b}{1-a} \right) \right\} = \cos \left\{ \pi + \frac{\pi}{\log a} \log \left(x - \frac{b}{1-a} \right) \right\} \\ &= -\cos \frac{\pi}{\log a} \log \left(x - \frac{b}{1-a} \right), = -\phi x, \end{aligned}$$

that is, $\phi x + \phi y = 0$.

If $b = 0$, then

$$y = ax \quad \text{and} \quad \phi x = \cos \frac{\pi}{\log a} \log x;$$

in fact, repeating the proof for this particular case,

$$\phi y = \cos \frac{\pi \log ax}{\log a} = \cos \pi \left(1 + \frac{\log x}{\log a} \right) = -\cos \frac{\pi \log x}{\log a}, = -\phi x,$$

that is,

$$\phi x + \phi y = 0.$$

Considering then (x, y) as the coordinates of a point on the curve $\Omega = 0$, we have, as above,

$$\chi\phi(x + iy) + \chi\phi(x - iy) = 0,$$

where ϕ is a real function determined as above, and χ is any real or imaginary odd function. This being so, assume

$$x_1 + iy_1 = e^{\chi\phi(x+iy)},$$

then also

$$x_1 - iy_1 = e^{\chi\phi(x-iy)},$$

and consequently

$$x_1^2 + y_1^2 = e^{\chi\phi(x+iy) + \chi\phi(x-iy)} = 1,$$

that is, we have the circumference of the circle $x_1^2 + y_1^2 - 1 = 0$ corresponding to the given curve $\Omega = 0$.

Suppose that the curve $\Omega = 0$ is a closed curve: and then writing

$$\xi + i\eta = \chi\phi(x + iy),$$

and therefore

$$\xi - i\eta = \chi\phi(x - iy),$$

we thence have

$$2\xi = \chi\phi(x + iy) + \chi\phi(x - iy),$$

a real function of (x, y) .

(1) Assume that it is possible to find χ , such that ξ as defined by this last equation shall be throughout the area of the curve $\Omega = 0$ finite and continuous, except only that in the neighbourhood of a given point, taken to be the point $x = 0, y = 0$, it is $= \log \sqrt{(x^2 + y^2)}$.

(2) At the boundary of the area $\Omega = 0$, ξ is $= 0$.

(3) Throughout the area, ξ satisfies the partial differential equation

$$\frac{d^2\xi}{dx^2} + \frac{d^2\xi}{dy^2} = 0.$$

These conditions being satisfied, the equation

$$x_1 + iy_1 = e^{\xi + i\eta},$$

that is,

$$x_1 + iy_1 = e^{\chi\phi(x+iy)},$$

gives an orthomorphosis of the area $\Omega = 0$ into the circle $x_1^2 + y_1^2 - 1 = 0$, the point $x = 0, y = 0$ corresponding to the centre of the circle; (2) and (3) are satisfied as above: it remains only to satisfy (1), viz. the function χ is determined not by any equation, but only by this condition as to finiteness and continuity; and if it be thus determined, then the foregoing equation $x_1 + iy_1 = e^{\chi\phi(x+iy)}$ gives the required orthomorphosis.

For instance, let the curve $\Omega = 0$ be the parabola $y^2 = 4(1 - x)$, which may be regarded as a closed curve bounding the infinite parabolic area. We have $2x = \xi + \eta$, $2iy = \xi - \eta$, whence the equation is

$$-\frac{1}{4}(\xi - \eta)^2 = 4 - 2\xi - 2\eta,$$

that is,

$$\xi^2 - 2\xi\eta + \eta^2 - 8\xi - 8\eta + 16 = 0,$$

whence $\sqrt{\xi} + \sqrt{\eta} - 2 = 0$, or writing this in the form

$$\frac{1}{2}\sqrt{\xi} + \frac{1}{2}\sqrt{\eta} - 1 = 0,$$

we have $\phi\xi = \sqrt{\xi} - 1$, and assuming that χ can be found so that the condition as to finiteness and continuity is satisfied, then the orthomorphosis is given by

$$x_1 + iy_1 = \exp \chi(\sqrt{\xi} - 1), = \exp \chi\{\sqrt{(x + iy)} - 1\}.$$

Assuming

$$\chi\omega = \frac{1}{2}i\pi\omega + \log \frac{1 - i \exp \frac{1}{2}i\pi\omega}{1 - i \exp(-\frac{1}{2}i\pi\omega)},$$

which is obviously an odd function, we have

$$\begin{aligned} \exp \chi\omega &= \exp \frac{1}{2}i\pi\omega \cdot \frac{1 - i \exp \frac{1}{2}i\pi\omega}{1 - i \exp(-\frac{1}{2}i\pi\omega)} = \frac{\exp \frac{1}{2}i\pi\omega - i \exp i\pi\omega}{1 - i \exp(-\frac{1}{2}i\pi\omega)} \\ &= \frac{\exp \frac{1}{2}i\pi\omega(1 - i \exp \frac{1}{2}i\pi\omega)}{1 - i \exp(-\frac{1}{2}i\pi\omega)}, \end{aligned}$$

which is

$$= \tan^2 \frac{1}{4}\pi(\omega + 1),$$

and hence, for ω writing $\sqrt{(x + iy)} - 1$, we have

$$x_1 + iy_1 = \exp \chi\{\sqrt{(x + iy)} - 1\}, = \tan^2 \frac{1}{4}\pi\sqrt{(x + iy)}.$$

This satisfies the required conditions as to finiteness and continuity; and in particular, we have

$$\xi + i\eta = \log \tan^2 \frac{1}{4}\pi\sqrt{(x + iy)},$$

so that, x and y being small,

$$\xi + i\eta = \log \frac{\pi^2}{16}(x + iy), \quad \xi - i\eta = \log \frac{\pi^2}{16}(x - iy),$$

that is,

$$\xi = \log \frac{\pi}{4}\sqrt{(x^2 + y^2)}.$$

Hence we have the known result: the orthomorphosis of the parabola $y^2 = 4(1 - x)$ into the circle $x_1^2 + y_1^2 - 1 = 0$ is given by the equation $x_1 + iy_1 = \tan^2 \frac{1}{4}\pi\sqrt{(x + iy)}$.

Consider the ellipse, where $a^2 - b^2 = 1$, or say

$$\frac{x^2}{\frac{1}{4}(M+\frac{1}{M})^2} + \frac{y^2}{\frac{1}{4}(M-\frac{1}{M})^2} = 1.$$

I show, by a less direct process, how to express this equation in the required form $\phi\xi + \phi\eta = 0$. In fact, writing

$$\xi = x + iy, \quad \eta = x - iy,$$

the equation of the ellipse is the rationalised form of

$$i\eta + \sqrt{(1 - \eta^2)} = M^2\{i\xi + \sqrt{(1 - \xi^2)}\}.$$

To show that this is so, call for a moment the right-hand side Ω , the equation is

$$\sqrt{(1 - \eta^2)} = \Omega - i\eta,$$

hence

$$\begin{aligned} 1 - \eta^2 &= \Omega^2 - 2\Omega i\eta - \eta^2, \\ 2\Omega i\eta &= \Omega^2 - 1, \end{aligned}$$

or

$$2i\eta = \Omega - \frac{1}{\Omega} = M^2\{i\xi + \sqrt{(1 - \xi^2)}\} - \frac{1}{M^2}\{i\xi + \sqrt{(1 - \xi^2)}\}^{-1},$$

therefore

$$2i\eta = \Omega - \frac{1}{\Omega} = M^2\{i\xi + \sqrt{(1 - \xi^2)}\} + \frac{1}{M^2}\{i\xi - \sqrt{(1 - \xi^2)}\},$$

that is,

$$-4\eta^2 = -4\xi^2(M^2 + \frac{1}{M^2})\xi^2 - (M^2 - \frac{1}{M^2})^2,$$

or say

$$-\xi^2 - \eta^2 + (M^2 + \frac{1}{M^2})\xi\eta - \frac{1}{4}(M^2 - \frac{1}{M^2})^2 = 0;$$

viz. substituting for ξ, η their values, this is

$$-2(x^2 - y^2) + (M^2 + \frac{1}{M^2})(x^2 + y^2) - \frac{1}{4}(M^2 - \frac{1}{M^2})^2 = 0,$$

that is,

$$(M - \frac{1}{M})^2 x^2 + (M + \frac{1}{M})^2 y^2 - \frac{1}{4}(M^2 - \frac{1}{M^2})^2 = 0,$$

or finally, it is

$$\frac{x^2}{\frac{1}{4}(M+\frac{1}{M})^2} + \frac{y^2}{\frac{1}{4}(M-\frac{1}{M})^2} - 1 = 0,$$

as it should be.

Starting then from the relation

$$i\eta + \sqrt{(1 - \eta^2)} = M^2\{i\xi + \sqrt{(1 - \xi^2)}\},$$

and writing

$$\phi\xi = \cos \frac{\pi}{2\log M} \log\{i\xi + \sqrt{(1 - \xi^2)}\},$$

we have

$$\begin{aligned} \phi\eta &= \cos \frac{\pi}{2\log M} \log\{i\eta + \sqrt{(1 - \eta^2)}\} \\ &= \cos \frac{\pi}{2\log M} \left[\log M^2 + \log\{i\xi + \sqrt{(1 - \xi^2)}\} \right] \\ &= \cos \left[\pi + \frac{\pi}{2\log M} \log\{i\xi + \sqrt{(1 - \xi^2)}\} \right] \\ &= -\cos \frac{\pi}{2\log M} \log\{i\xi + \sqrt{(1 - \xi^2)}\}, = -\phi\xi, \end{aligned}$$

that is, we have

$$\phi\xi + \phi\eta = 0,$$

as the required transformation of the equation of the ellipse

$$\frac{x^2}{\frac{1}{4}(M+\frac{1}{M})^2} + \frac{y^2}{\frac{1}{4}(M-\frac{1}{M})^2} = 1.$$

We hence derive the known formula for the orthomorphosis of the ellipse into the circle $x_1^2 + y_1^2 - 1 = 0$.